EE446 Assignment 1
Asaf Iron-Jobes
Part 1
E)
Further reduction is possible. I combined knowledge distillation with int8 quantization which brought the overall model size to 3.68KB which is smaller than any previous technique. The smallest previous model was 5.82KB. Pruning was the least effective compression technique reducing the size from 14.18 to 14.14.
The key to this approach I took is that knowledge distillation and quantization are orthogonal in that one reduces the size of the model structure while the other reduces the size per neuron/weight. The result was a model with 3.68KB in size and 96% test accuracy. This slight drop from 100% is a fair tradeoff for a 4x size reduction in TinyML applications.

HW2 Part 2

1. [Dis] (5 pts) What types of "Learning Blocks" are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self-supervised learning, etc.)?
For each type, provide at least one example model that can be implemented using the free tier.

The free tier of edge impulse offers multiple learning blocks.

Supervised classification has an option for transfer learning which fine tunes the MobileNet architecture.

Supervised regression offers the Keras block. An example this can be used for is temperature estimation from multiple sensor measurements.

Unsupervised has anomaly detections options for models. One option is K-means clustering for anomaly detection. Another model for this could be a gausian mixture model. This could be used for an image model that check for anomaly detection for weather such as storms as an example.

2. [Dis] (5 pts) List all the types of layers that can be used under the "Classifier" section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

Dense
Fully connected layer, the simplest form of a neural network layer. Use this for processed data, such as th

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1D Convolution / pooling
Learn features that take spatial information into account along a single dimension. Use this for raw data, o

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2D Convolution / pooling
Learn features that take spatial information into account along two dimensions. Use this for raw data, or fo

Reshape
Turn one-dimensional data from a DSP block into multi-dimensional data. Use this as an input to a convol

Flatten
Flatten multi-dimensional data into a single dimension. You need to flatten data from a convolutional laye

Dropout
Reduce the risk of a model overfitting your dataset by randomly cutting a fraction of network connections

This is found from edge impulse website.

3. [Dis] (5 pts) What types of model compression options are available under the "Deployment" section in the free-tier Edge Impulse? List all available methods.

The deployment section in edge impulse has two model optimization options in the free tier.

Unoptimized (float 32) is the baseline which does not perform quantization on the model and keeps the weights as 32-bit values.

The quantized (int8) version does post-training quantization and converts the weights and activations from float to int8. This compresses the model significantly.

4. [Dis] (5 pts) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.

Edge impule has 3 public synthetic data generation blocks available at the free tier.

For Images they have black forest labs flux-pro synthetic image generator. This offers high quality images with relatively strong prompt following, detail, and diversity. This requires a replicate API key.

The Dall-E 3 synthetic image generator uses open Ais model to generate images from text and also requires an API key.

For synthetic audio data they have whisper synthetic voice generator. This is OpenAIs TTS model and can generate synthetic voice sample for use cases like KWS.

Synthetic data is important because you can get more data to train your model without having large datasets that are for your specific application. It can especially be useful to fill gaps in areas of datasets for underrepresented data points or application specific data needs. It can also eliminate manual labeling with can be expensive and take a long time.